

ProDiag AI V2

Machine Learning Evaluation Report

Agentic Predictive Maintenance Copilot for Industrial Assets

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This report summarizes the predictive maintenance ML experiments, model comparison, feature analysis, threshold analysis, and health score regression evaluation performed for ProDiag AI V2.

1. Experiment Overview

The ProDiag AI V2 ML pipeline evaluates supervised models for failure prediction and machine health-score estimation. Isolation Forest remains part of the anomaly-detection layer.

Component	Purpose
Failure Prediction	Predict whether failure occurs within the next 1 hour
Health Score	Estimate continuous machine health from 0–100
Anomaly Detection	Detect unusual sensor behaviour using Isolation Forest
Feature Analysis	Evaluate sensor, trend and equipment features
Threshold Analysis	Select an operating threshold for failure prediction

2. Failure Prediction Model Evaluation

Three supervised classification approaches were evaluated: Logistic Regression, Random Forest and XGBoost. The primary model-selection metric was F1-score, with recall treated as especially important because missed failures can represent significant maintenance risk.

	accuracy	precision	recall	f1	roc_auc	pr_auc
	0.9968	0.9896	0.9952	0.9924	0.9999	0.9996
t	0.9967	0.9842	1.0	0.992	1.0	1.0
ssion	0.9818	0.9991	0.9133	0.9543	0.9998	0.9991

Selected failure model: XGBoost

XGBoost achieved the highest F1-score in the evaluated comparison, with strong precision, recall and ranking performance.

3. Health Score Regression Evaluation

Health-score estimation was evaluated as a regression problem. Linear Regression, Random Forest and XGBoost were compared using MAE, RMSE and R².

model	mae	rmse	r2
XGBoost Regressor	3.3622	9.8492	0.9268
Random Forest Regressor	3.4736	9.1261	0.9372
Linear Regression	4.9731	11.8367	0.8943

Selected health-score model: XGBoost

XGBoost achieved the strongest evaluated regression performance across the primary error and goodness-of-fit metrics.

4. Feature Importance & Explainability

XGBoost feature importance and SHAP analysis were performed to understand which sensor and trend features contributed most to the model's predictions.

feature	importance	importance_percent
numeric__vibration_mm_s_rolling_mean_10	0.2307	23.0735

feature	importance	importance_percent
categorical__equipment_type_gearbox	0.1414	14.139
categorical__equipment_type_conveyor	0.1309	13.0935
numeric__vibration_mm_s	0.118	11.8004
numeric__rpm_rolling_mean_10	0.0877	8.7659
categorical__equipment_type_pump	0.0623	6.2338
numeric__rpm	0.0527	5.2654
numeric__temperature_c	0.0506	5.0572
numeric__current_a	0.043	4.2954
numeric__temperature_c_rolling_mean_10	0.0284	2.8441

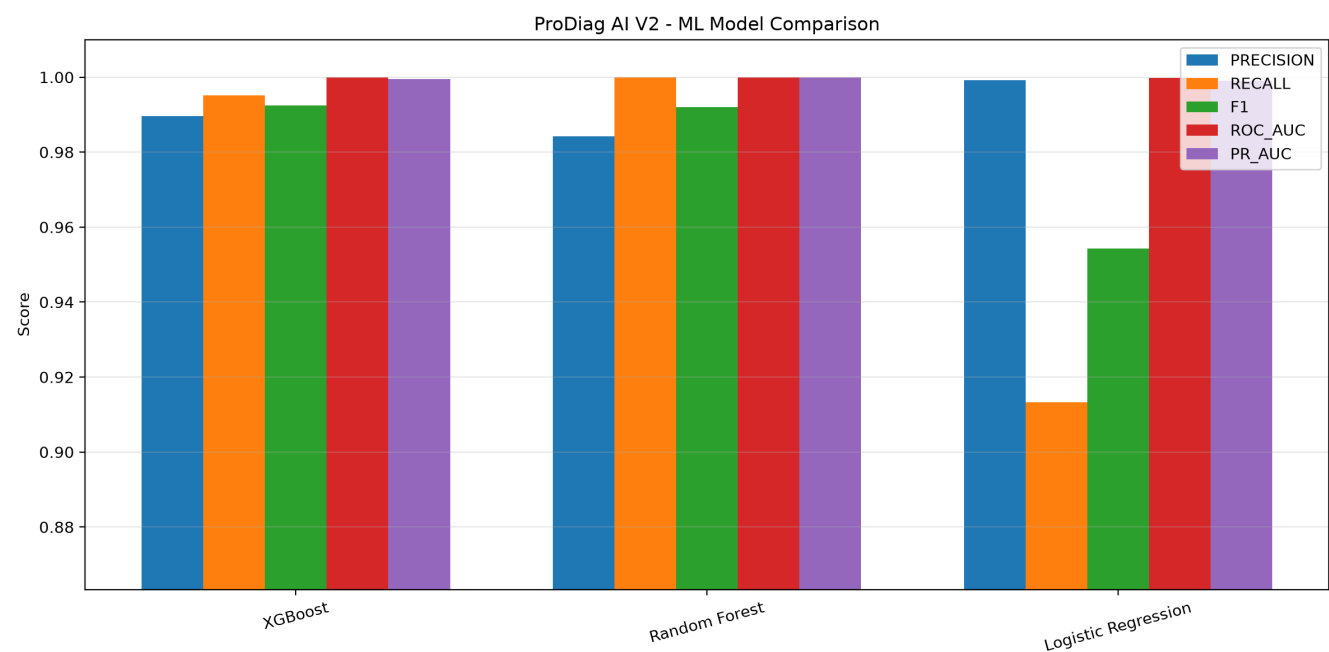
5. Failure Prediction Threshold Analysis

Multiple probability thresholds were evaluated to understand the precision/recall trade-off. The operating threshold should be treated as an engineering decision rather than a universal ML default.

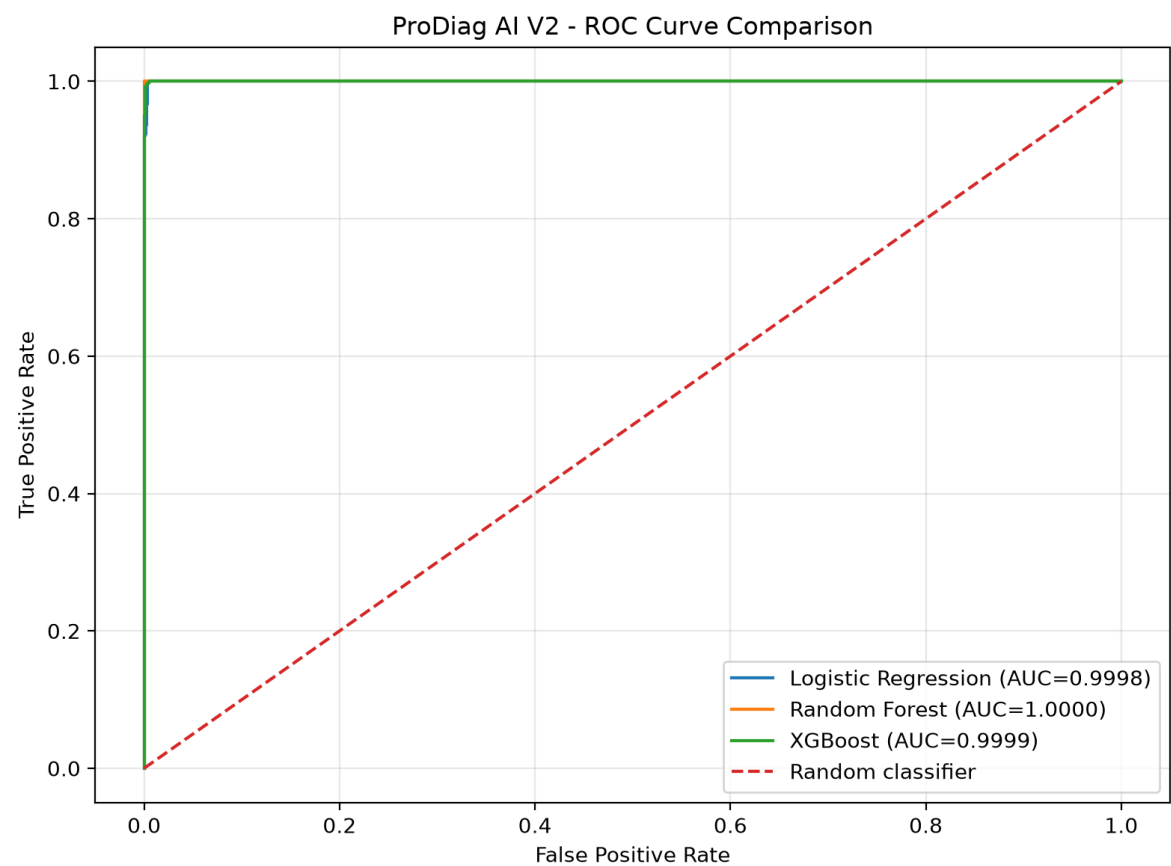
threshold	accuracy	precision	recall	f1	true_negatives	false_positives	false_negatives	true_positives
0.2	0.9927	0.9659	1.0	0.9826	4710.0	44.0	0.0	1246.0
0.3	0.995	0.9765	1.0	0.9881	4724.0	30.0	0.0	1246.0
0.4	0.9958	0.9834	0.9968	0.99	4733.0	21.0	4.0	1242.0
0.5	0.9968	0.9896	0.9952	0.9924	4741.0	13.0	6.0	1240.0
0.6	0.9968	0.9904	0.9944	0.9924	4742.0	12.0	7.0	1239.0
0.7	0.9973	0.9952	0.992	0.9936	4748.0	6.0	10.0	1236.0
0.8	0.995	0.9959	0.9799	0.9879	4749.0	5.0	25.0	1221.0

6. Evaluation Visualizations

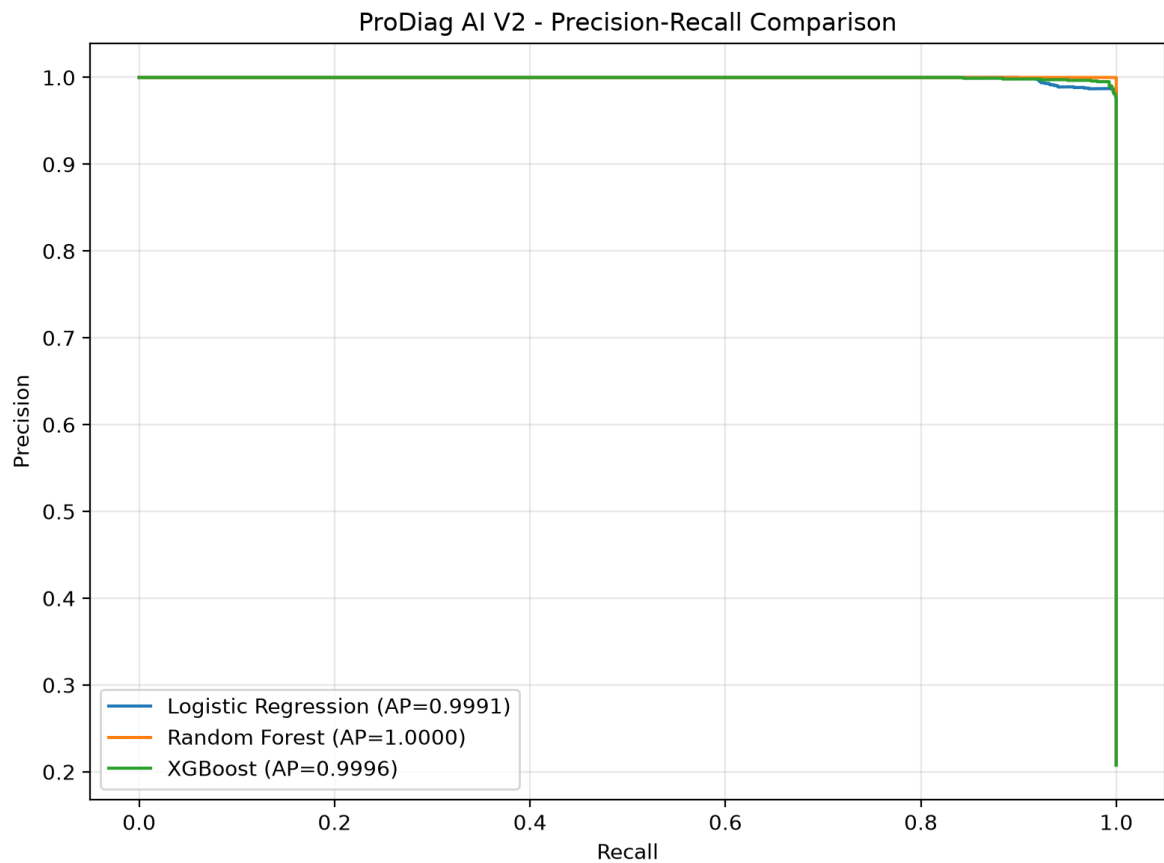
Model Comparison



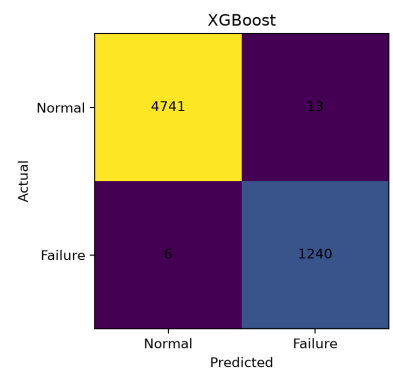
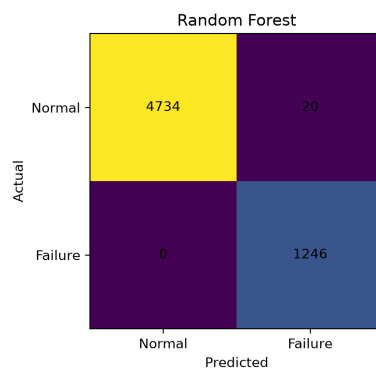
Roc Curve Comparison



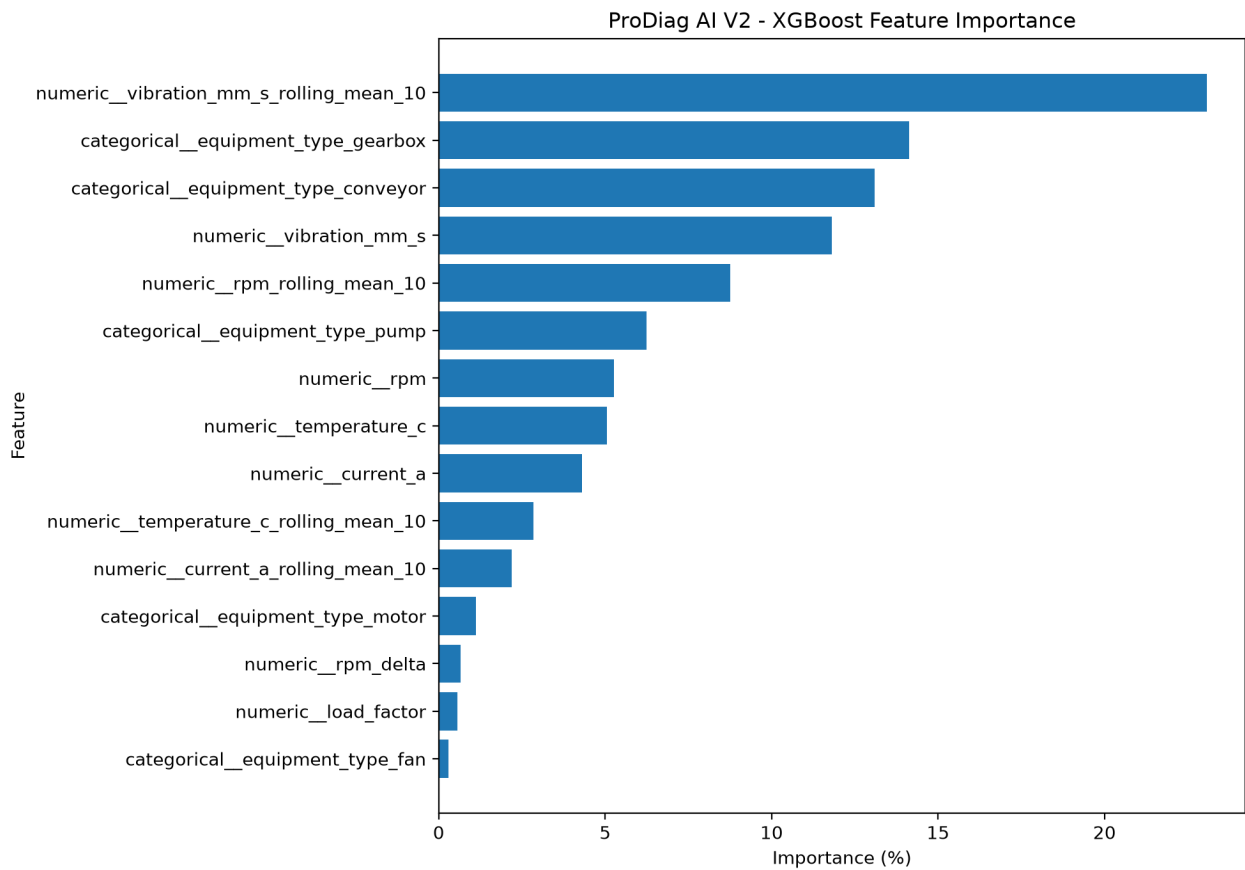
Precision Recall Comparison



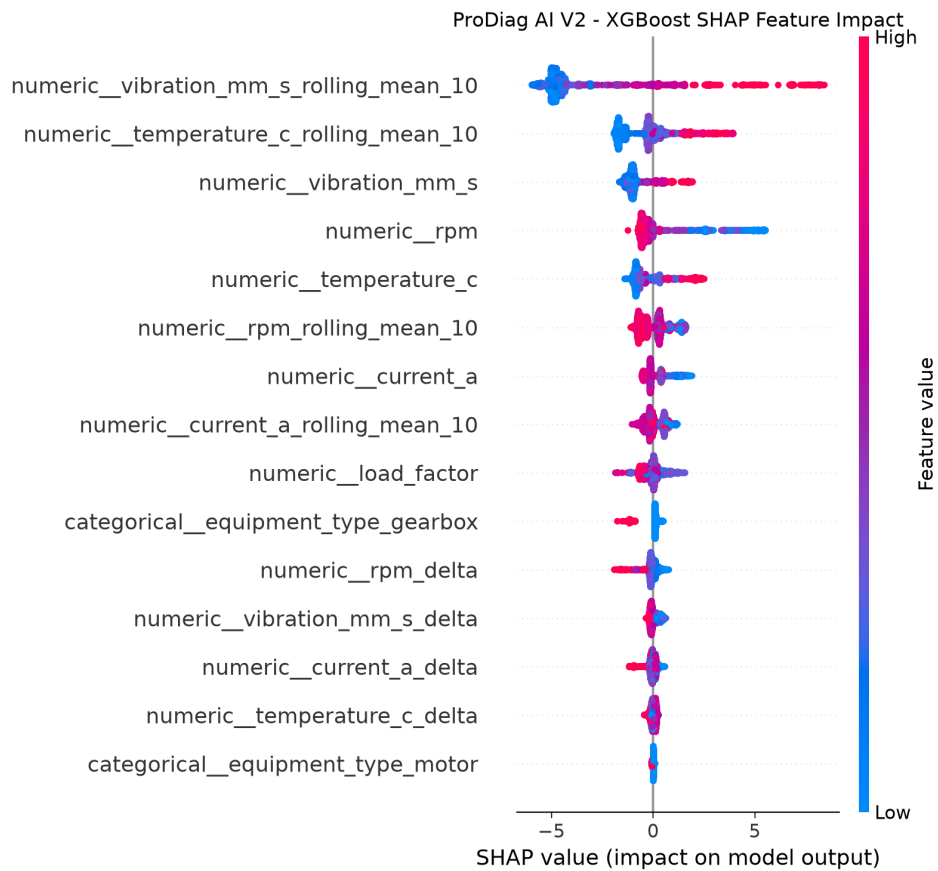
Confusion Matrices



Xgboost Feature Importance

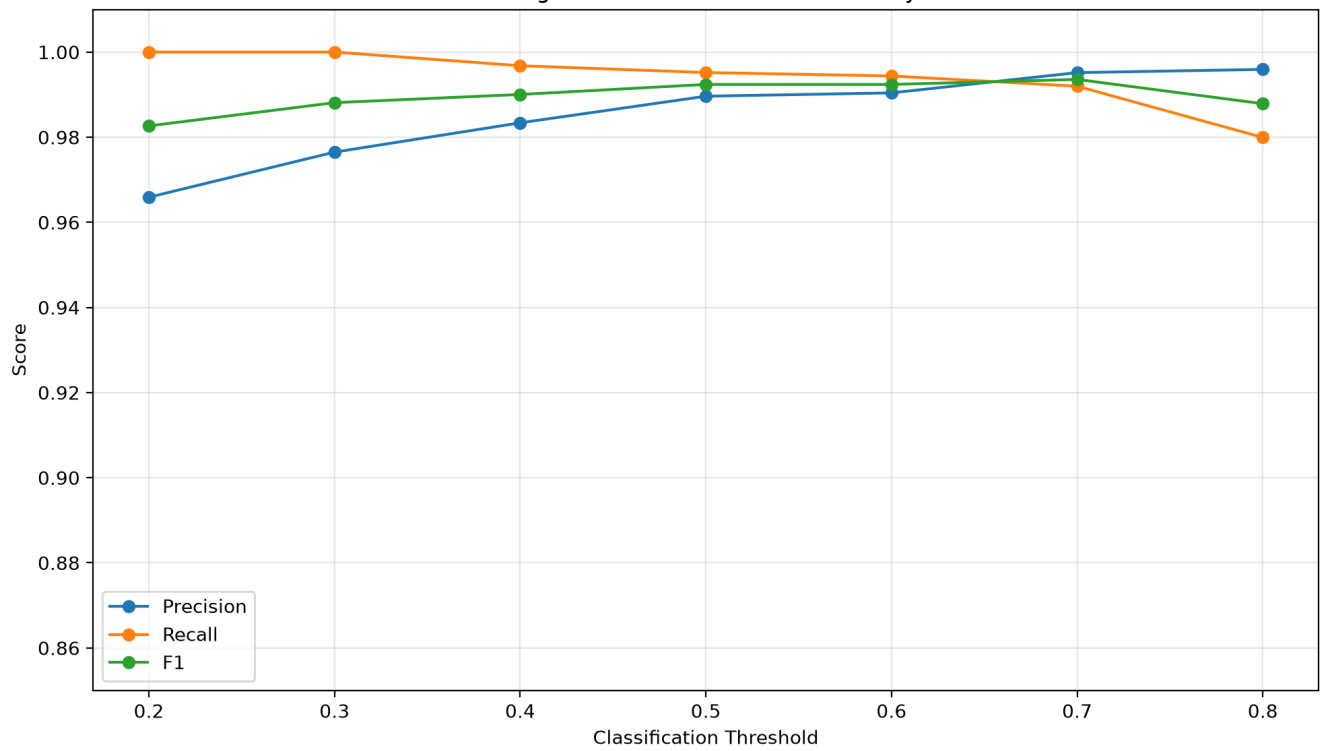


Xgboost Shap Summary

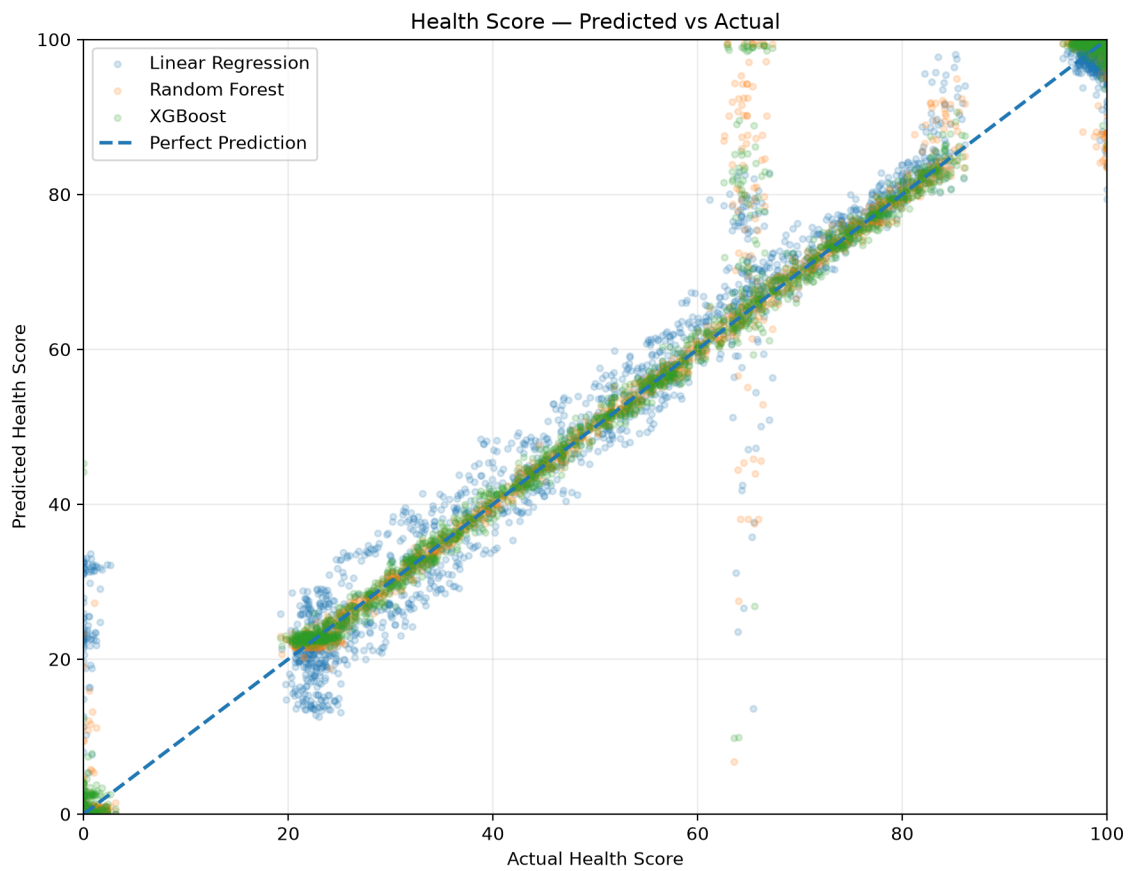


Xgboost Threshold Analysis

ProDiag AI V2 - XGBoost Threshold Analysis



Health Score Predicted Vs Actual



Health Score Residual Distribution

Health Score — Residual Distribution

